Wk7.2 Bagging and boosting

Thursday, October 22, 2026 9:17 AM

High level ideas & basic calcs,

Exam 1 Summary (Fill in the key topics!)

3-4 Key def question from list of lects.

Ch1 Intro – Very short intro (no topics)

Ch2 statistical learning – Thm ~~Bias-variance tradeoff~~, Thm reduc. & irreduc. error, Regression vs classif, superv vs unsup, key topic overview — prob review + expectation

$Y = f(x) + \varepsilon$

Ch3 Linear regression – SLR, MLR, OLS of $\hat{\beta}_i$ fitting, metrics RSE, RSS, R^2, pred vs inference, CI, HT.

Ch4 Classification – $p \to \frac{p}{1-p} \to \ln\left(\frac{p}{1-p}\right)$ (odds) $\to$ LR, Logistic regression, ~~LDA, QDA, NB (Bayes thm)~~, classif metrics

Ch5 Resampling Methods – Bootstrap for SE approx (CI & HT) + CV to approx test error

Ch6 Linear model selection & regularization (shrinkage) – variable sel. to prev over-fitting, Ridge $\|\cdot\|_2$ + Lasso $\|\cdot\|_1$ — Elastic net avgs 2.

Ch 8 Tree-based models.

Decision tree basics — Regression case ✓

$Y = f(X) + \varepsilon$

Train data: Partition predictor space & approx $y \approx \hat{y} = \bar{y}_{R_j}$ (region avg) in best way (minimizes RSS) & iterate (until rule met or CV/shrinking)

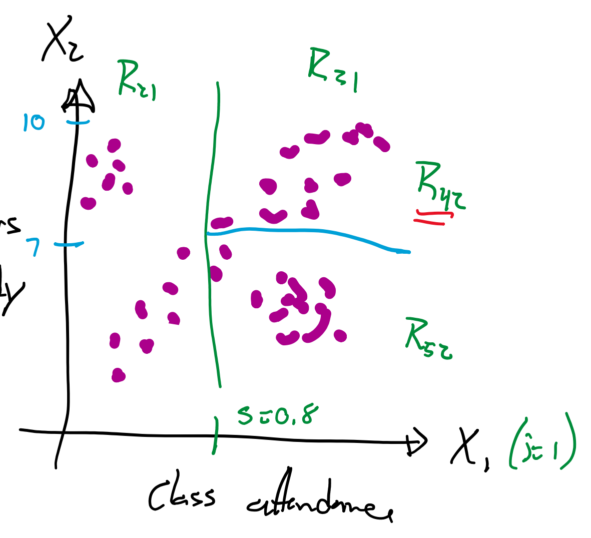
Almost same for classification. Key difference is metrics for splitting.

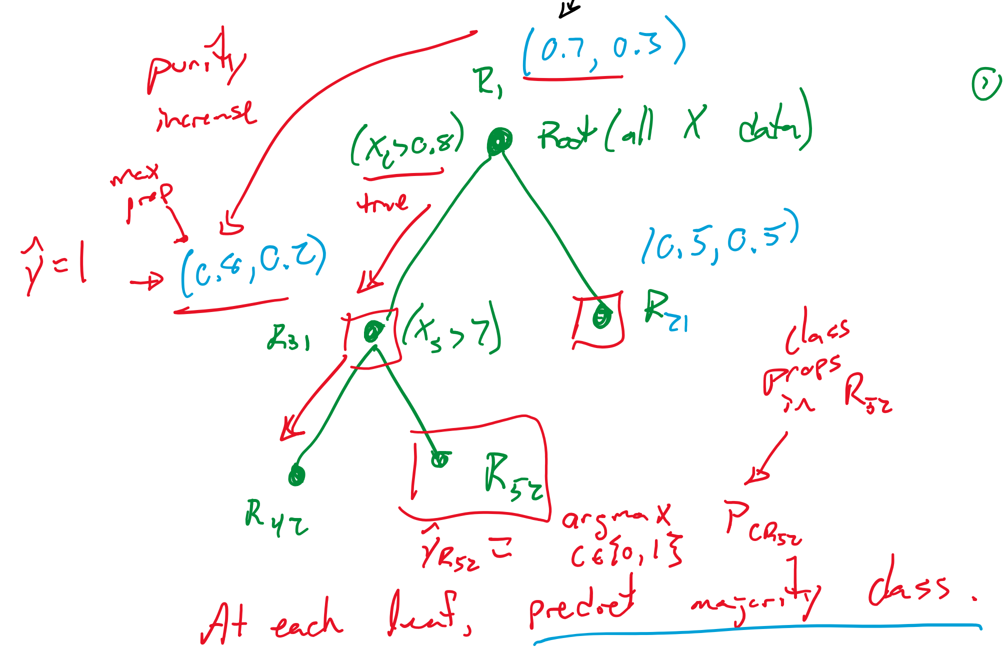
Decision tree for classification

$Y = f(X) + \varepsilon$

$Y \in \{$Pass exam, Fail exam$\}$ — 1, 0

$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_c \\ x_s \end{bmatrix}$ pass rate.





At each leaf, predict majority class.

① Consider all partitions of R_1

$R_{2j} = \{X \mid X_j > s\}$
$R_{3j} = \{X \mid X_j < s\}$ (all j & s)

How to chose best partition (X_j, s)? Separate classes optimally

3 ways: ① Classification error $= 1 - \max_{k \in \text{classes}} \hat{P}_{mk}$ or weighted (by sample) avg across 2 new regions (poor for class imbalance & 3+ classes)

$\hat{P}_{mk} =$ prop of k in m — node m, class k

☆ ② Gini coefficient: $G_m = \sum_k \hat{P}_{mk}(1 - \hat{P}_{mk})$ (avg of all props)
$= 1 - \sum_k \hat{P}_{mk}^2$ ~~total~~ variance-ish.

↳ Connects to $SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$ (Stat 145) & R^2 ish.

③ Entropy (information theory) $= -\sum_k \hat{P}_{mk} \ln(\hat{P}_{mk})$ measures disorder.

For any decision tree, get feature importance by reporting which variables give most improvements.

↳ Regression: RSS decrease.

↳ Classification: Purity increase.

DT advantages & disadvantages

Adv: Highly flexible (low bias) & interpretable.

Disadv: High variance ☹ ⟶ Low performance in general.

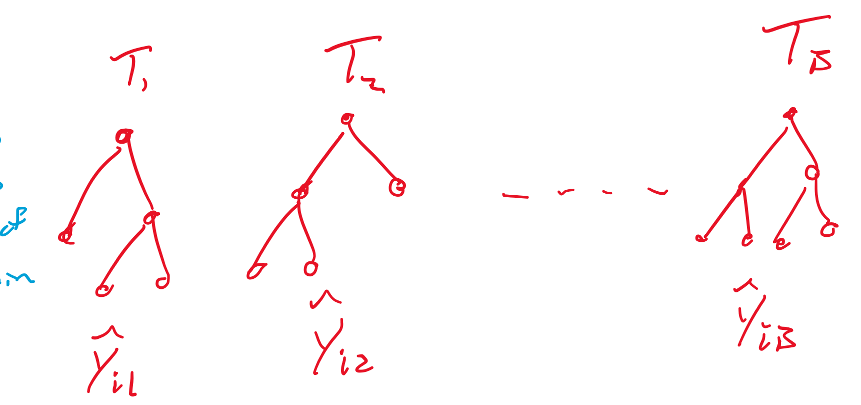
Solve by building many DTs & average predictions (Ensembling) gives low bias & low variance.

2 main approaches: ① Bootstrap train data (Bagging, bootstrap aggregate; ☆ Random forest — DT)
② Recursively correct predictions (☆ DT Boosting)
Any SL. ⟶

Bagging Build B DTs on bootstraps (B: sample n points from train data with replacement) of train data (size n).

1 2 … 10

1B is 2/3 of train



$\longrightarrow \hat{Y} = \begin{cases} \text{avg for regression} \\ \text{majority class for classify (props of classes give certainty)} \end{cases}$

OOB$_2$ - - - → OOB$_B$

BSS leaves out some of train set, call OOB (out of bag) ↑ ~1/3 train sample left out

→ Approx test error as avg OOB error

Issue: If n large, most 1B data is very similar & $T_1, \dots, T_B$ are almost the same.

Random forest fixes this by only using some predictors $X = (X_1, \dots, X_p)$, subset of $m \le p$.

$m = p$ (Bagging)
☆ $m \approx \sqrt{p}$ (standard)
$m \approx p/2$
⋮